Mofe Omolade

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Education

University of British Columbia — Vancouver, BC
Bachelor of Applied Science in Computer Engineering — *Expected May 2029*

Technical Skills

Tools: Altium, KiCAD, LTspice, STM32CubeIDE, Quartus Prime, Questa, Git
Languages: C/C++, Python, Assembly, MATLAB, SystemVerilog, Java
Embedded Systems: ESP32, STM32, CAN, I2C, SPI, isoSPI, UART, USB
Lab: Oscilloscope, Logic Analyzer, Digital Multimeter, Function Generator, Soldering, Rework

Experience

Electrical Team Member — September 2026 – Present
UBC Formula Electric — *Vancouver, BC*

- Working with a multidisciplinary team of 100+ students to design, build, and race an FSAE Electric race car
- Executing schematic and layout revisions for BMS daughterboard PCBAs across a 588V tractive accumulator, integrating the ADBMS6830B battery cell monitor to measure pack voltage, temperature, and balance cells
- Transitioning cell balancing from internal ADBMS switching to discrete external MOSFETs, increasing balancing current to 300 mA to reduce pack equalizing time by 33%
- Performing SPICE simulations to validate ADMBS open wire check behaviour and quantify P-channel MOSFET quiescent leakage currents within cell buffer circuitry
- Leading PCBA bring-up and hardware verification across new revisions, validating power rails, high-voltage isolation, and isoSPI communication integrity under varying accumulator loads

Electronics Engineering Intern — May 2026 – August 2026
Fulcrum Air — *Calgary, AB*

- Owned the complete design life cycle of an 8-layer 4-channel motor controller PCBA to drive 12V actuators at 3A, replacing off-the-shelf Pololu drivers to improve reliability and integration ease for future robot developments
- Designed system architecture around Teensy 4.1 and IFX007T half bridges, supporting multichannel motor control, CAN communication, analog current sensing, and sub-millimeter Hall-effect positioning
- Implemented orthogonal routing, dedicated power and ground planes, and ground stitching vias to keep PCB surface temperature rise below 25°C under continuous 150 W draw
- Engineered a power hold-up circuit and falling-edge ISR using a voltage divider and 1000µF bulk capacitor, enabling EEPROM backups of actuator positioning data within 5 ms of power disconnection
- Built a test bench and developed firmware to validate PCB function, verifying MCU pinouts, voltage regulation, inrush current stabilization, CAN signal integrity, and bidirectional actuation under dynamic loads

Projects

5V Buck Converter | *LTspice, Altium* — August 2026 – Present

- Designed an asynchronous buck converter PCBA stepping down an 18V–25.2V input to 5V at up to 500 mA with 85% peak efficiency, maintaining CCM down to a 100 mA minimum load
- Developed an active Type III compensation network using an MCP6021 Op-amp to cancel the 1.57 kHz LC double pole, achieving a 61.8° phase margin and 22.3 dB gain margin at a 3.26 kHz crossover frequency
- Implemented an analog voltage mode PWM controller at 100 kHz, driving an LM334 constant-current source into a TLC555 timer to generate a linear sawtooth ramp fed into an MCP6561 comparator
- Engineered RC soft-start circuit on the 2.5V reference to suppress overshoot and limit inrush current to 10 mA
- Routed a compact 4-layer PCB, isolating sensitive feedback and timing traces from the high di/dt switching node
- Validated power rail voltages with DMM and tested step-load transient response using an oscilloscope

Self-Balancing Robot | *C/C++, ESP32, I2C* — July 2026 – August 2026

- Developed drivers for the MPU-6050 IMU sensor to parse gyroscope and accelerometer data over I2C
- Implemented non-blocking sensor data acquisition and applied a complementary filter to fuse accelerometer and gyroscope readings, achieving low-latency angle estimation while reducing gyroscope angular drift by 85%
- Engineering a custom closed-loop PID controller from first principles without third-party frameworks, incorporating anti-windup clamping, derivative filtering, and deadband compensation
- Programmed PWM generation using ESP32-C3 hardware timers to drive L298N dual-motor bridges